

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	Slot - A	Date:	24/07/26

Code of Conduct

I K. Sujith (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Sujith Sai Vighnesh

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

88
100

$$\text{Total Spectrum} = 45 \text{ MHz} \approx 45,000 \text{ kHz}$$

$$\text{Channel} = 25 \Rightarrow \text{Full} \therefore 2 \times 25 = 50$$

$$\text{Channels} = \frac{45,000}{50} \Rightarrow 900$$

$$\begin{aligned} \text{Channels per cell} &= \text{Total channels} / N \\ &= 900 / 7 \\ &= 128.57 \end{aligned}$$

$$\begin{aligned} \text{Total simultaneous voice channels} &= \text{Carriers} \times \text{Slots per carrier} \\ &= 100 \times 8 \\ &= 800 \end{aligned}$$

$$\begin{aligned} \text{Power Ratio} &= 80/4 \Rightarrow 20 \\ \text{in dB} &= 10 \log(20) \approx 13.01 \text{ dB} \\ &\approx 13 \text{ dB} \end{aligned}$$

$$\begin{aligned} \text{C/I} &= \text{Signal} - \text{Interference} \\ &= -60 - [-92] \Rightarrow -60 + 92 \\ &= 32 \text{ dB} \end{aligned}$$

$$\begin{aligned} \text{Total channels} &= \frac{42,000 \text{ kHz}}{300 \text{ kHz}} = 140 \\ \text{Per cell} &= 140 / 7 = 20 \end{aligned}$$

$$\text{Received Power} \propto 1/d^n$$

$$\text{Doubling Power} = 2^4 = 16$$

$$\Rightarrow \frac{16 \times 4}{164} = 4$$

8)

$$n = 2$$

$$\text{Doubling Radius} = 2^2 = 4$$

$$\Rightarrow 4 \text{ Times}$$

9)

$$\text{Cluster size} = 4$$

$$\text{Per cell} = 15 \Rightarrow 4 \times 15 = 60$$

10)

$$v = 100 \text{ km/h}$$

$$\text{Doppler Shift} = v \times f/c$$

$$= 27.78 \times 900 \times 10^6 / 3 \times 10^8$$

$$= 83.3 = 83 \text{ Hz}$$

11)

$$\text{Slot duration} = \frac{\text{Frame Time}}{\text{Slots}}$$

$$= \frac{8}{5} = 0.625 \text{ ms}$$

12)

$$\text{Carriers} = \frac{\text{Total spectrum}}{\text{spacing}} = \frac{120}{\frac{24000 \text{ KHz}}{800 \text{ KHz}}}$$

$$= 120 \text{ KHz}$$

$$\text{Total capacity} = \text{cells} \times \text{calls}$$

$$= 250 \times 6$$

$$\text{Total capacity} = \text{cells} \times \text{calls}$$

$$= 250 \times 6$$

$$= 1500$$

16)

$$\text{Active calls supported} = \text{arrival time} \times \text{avg duration}$$

$$= 4000 \times 180 = 720,000$$

$$\text{Total band width} = \text{channels} \times \text{bandwidth}$$

$$= 150 \times 25$$

$$= 3750 \text{ KHz}$$

$$= \cancel{150} \Rightarrow 3750 \text{ KHz}$$

$$\text{wasted bandwidth} = 29 \times 40$$

$$= 1160 \text{ KHz}$$

$$= 1160 \text{ KHz}$$

$$\text{Speed} = 90$$

$$D = 1.5 \times 2 = 3.0 \text{ km} \approx 3000 \text{ m}$$

$$\text{Time per cell} = 3000 / 25$$

$$= 120$$

$$\text{Per hour} = \frac{3600}{120} = 30$$

$$\text{Total users} = \text{Centers} \times \text{base Stations} \times$$

$$= 8 \times 40 \times 250$$

$$= 80,000$$

$$\text{Channels} = \frac{\text{Total Spectrum}}{\text{Channel width}} = \frac{6000}{30} = 200$$

$$\text{Total Call-minutes} = 1200 \times 4 \times 2.5$$

$$= 12,000$$

$$\text{Traffic} \Rightarrow \frac{12,000}{60} \Rightarrow 200$$

$$\frac{\text{Total band width}}{\text{Bandwidth per user}} = \frac{4000}{400} = 10$$

25)

$$\text{Capacity} = \text{Users Per cell} \times \text{Cells}$$

$$= 600 \times 1200$$

$$= 720000$$

26)

$$\text{Paging Operations} = \text{Total cells} / \text{Cells per Operation}$$

$$= \frac{200}{4} = 50$$

27)

$$\text{Offered Traffic} = \text{arrival rate} \times \text{Avg Duration}$$

$$= 40000 \times 3 = 120000$$

$$\text{Over Capacity} = 120000 - 100000$$

$$= 20000$$

$$\text{Blocked} = \frac{20000}{120000} \approx 0.17$$

28)

$$\text{Roaming Users} = \frac{15}{100} \times 60000$$

$$= 9000$$

29)

$$\text{Utilization} = \frac{\text{Peak Traffic}}{\text{Processing Rate}} = \frac{9000}{12000} = 0.75 \approx 75\%$$

30)

$$\text{Total Channels} = \frac{\text{Band width}}{\text{Channel width}} = \frac{48000 \text{ kHz}}{200} = 240$$

$$\text{Chan Per cell} = \frac{240}{4} = 60$$

31)

$$\text{Channels} = \frac{\text{Total spectrum}}{\text{Channel width}} = \frac{30000}{200} = 150$$

$$\text{Successful Hand off} = 99\% \times 2500$$

$$= \frac{99}{100} \times 2500 \Rightarrow 2475$$

$$\text{Total Capacity} = \frac{\text{Users}}{\text{Cells per Cell}} = \frac{200}{7} \Rightarrow$$

$$\begin{aligned} \text{Total Capacity} &= \text{Users} \times \text{Cells} \\ &= 900 \times 7 \\ &= 6300 \end{aligned}$$

$$\begin{aligned} \text{Slot Duration} &= \text{Frame Time} / \text{Slots} \\ &= 4.8 / 8 \\ &\Rightarrow 0.6 \text{ ms} \end{aligned}$$

$$v = 150 \text{ km/hr}$$

$$\begin{aligned} \text{Offered traffic} &= 70,000 \times 4 \\ &= 280,000 \end{aligned}$$

$$\begin{aligned} \text{Excess} &= 280,000 - 250,000 \\ &= 30,000 \end{aligned}$$

$$\text{Fraction Blocked} = \frac{30,000}{280,000} \approx 0.11$$

$$\begin{aligned} \text{Total Users in cluster} &= \text{Users} \times \text{Cells} \\ &= 600 \times 9 \\ &= 5400 \end{aligned}$$

$$\begin{aligned} \text{Call dropped / sec} &= 0.4\% \times 60,000 \\ &= \frac{4}{100} \times 60,000 \\ &\Rightarrow 2400, \end{aligned}$$

$$\begin{aligned}
 40) \quad \text{Path loss} &= T \times \text{Power} - R \times \text{Power} \\
 &= 43 - (-87) \\
 &= 130 \text{ dB}
 \end{aligned}$$

$$\begin{aligned}
 41) \quad \text{Total Coverage} &= \text{Coverage no of stations} \\
 &= 25 \times 150 \\
 &= 375
 \end{aligned}$$

$$42) \quad \text{Total channels} = \frac{48000}{200} = 240 \Rightarrow 240/6 = 40$$

$$43) \quad \text{Paging cycles} = \frac{\text{Total users}}{\text{Users per cycle}} = \frac{150000}{6000} = 25$$

$$44) \quad \text{Handoff per call} = \frac{\text{Call duration}}{\text{Handoff interval}} = 9/3 = 3$$

$$\begin{aligned}
 45) \quad \text{SNR} &= \text{Signal} - \text{Noise} \\
 &= 85 - (-115) = 30 \text{ dB}
 \end{aligned}$$

$$\begin{aligned}
 46) \quad \text{Min bandwidth} &= \text{Users} \times \text{rate} \\
 &= 250 \times 150 = 37500
 \end{aligned}$$

$$\begin{aligned}
 47) \quad \text{Total call} &= \text{Users} / \text{rate} \times \text{Calls} \times \text{Duration} \\
 &= \frac{150000}{150} \times 4 \times 2.5 \\
 &= 15000 \text{ min}
 \end{aligned}$$

$$\text{Traffic} = \frac{15000}{60} = 250$$

$$\begin{aligned}
 48) \quad \text{Bandwidth} &= \text{Channels} \times \text{bandwidth} \\
 &= 250 \times 12 \\
 &= 3000 \text{ KHz}
 \end{aligned}$$

$$\begin{aligned}
 49) \quad \text{Total throughput} &= \text{Connections} \times \text{rate} \\
 &= 1500 \times 600 = 900000 \text{ Kbps} = 900 \text{ Mbps}
 \end{aligned}$$